

RUTHRESH D.R

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SUMMARY

Mechanical Engineering student with hands-on experience in **mechanical design, 3D CAD modelling, engineering drawings, and FEA**. Proficient in **SolidWorks, Creo, Fusion 360, and ANSYS Workbench** with practical experience in **part and assembly design, weldments, sheet metal, DFM/DFA, meshing, and structural analysis**. Experienced in translating engineering concepts into manufacturable designs through internships and academic projects. Seeking an internship opportunity.

EDUCATION

B.E. Mechanical Engineering	2023–2027
Velammal Engineering College, Anna University	CGPA: 8.31 upto 6th semester
Higher Secondary	2023
Dharani Matric Higher Secondary School	79%

TECHNICAL SKILLS

CAD Software: SolidWorks, Creo, Fusion 360

FEA / CAE: ANSYS Workbench, Static Structural Analysis, Modal Analysis, Meshing, Fluent CFD

Design: 3D Part Modelling, Assembly Design, Weldments, Sheet Metal, Engineering Drawings, GDT, DFM, DFA

Engineering: Mechanical Design, Machine Design, Material Selection, Structural Analysis, Design Validation, Manufacturing Processes

Programming: Python, C, C++

PROJECTS

PCB Milling Machine 2026

- Designed and fabricated a compact CNC-based PCB milling machine using SolidWorks and lead screw mechanisms.
- Developed the machine frame, component mounting, and linear motion arrangement considering rigidity, alignment, and manufacturability.
- Selected NEMA17 motors, T8 lead screws, linear guides, and structural components based on design requirements.
- Improved structural rigidity and reduced vibration through iterative CAD-based design modifications.
- Integrated Arduino-based motion control with the mechanically developed system for automated machining.

Fixed-Wing UAV – SAE DDC 2026

- Designed lightweight structural components and assemblies with focus on **strength-to-weight ratio and manufacturability**.
- Developed CAD models for the fuselage, wing structure, ribs, spars, and supporting components.
- Performed **FEA and CFD analysis** to evaluate structural performance and aerodynamic behaviour.
- Selected suitable materials and structural members based on loading conditions and design requirements.
- Participated in fabrication, assembly, testing, and iterative design improvement of the prototype.

Autonomous Multirotor UAV – SAE ADDC 2025

- Designed the mechanical frame and structural components considering payload capacity, rigidity, and weight constraints.
- Developed and refined CAD models for integration of structural, propulsion, and electronic subsystems.
- Performed structural analysis to evaluate component strength under payload and operating loads.
- Supported fabrication, assembly, and prototype testing through iterative design modifications.

ANSYS Billboard FSI Analysis 2026

- Performed one-way Fluid-Structure Interaction analysis using **ANSYS Fluent and Static Structural**.
- Developed the computational domain, generated meshes, applied wind loading, and transferred pressure loads to the structural model.
- Evaluated structural deformation and stress response of the billboard under simulated wind loading.

EXPERIENCE

Design Engineer Intern – RK Invent Weld Automation Pvt. Ltd. 2025

- Designed mechanical components and assemblies for industrial automation equipment using **SolidWorks**.
- Developed **3D CAD models, weldments, sheet metal components, and manufacturing drawings**.
- Coordinated with fabrication teams to verify dimensions, manufacturability, and assembly feasibility.
- Applied **DFM principles** to simplify component fabrication and improve production efficiency.

Student Trainee Intern – Enexio Power Cooling Solutions India Pvt. Ltd. 2026

- Gained practical exposure to fabrication and manufacturing of Air-Cooled Condensers, cooling towers, and industrial heat-transfer equipment.
- Observed **welding, machining, assembly, hydro testing, quality inspection, and material handling** processes.
- Studied engineering drawings, manufacturing workflows, quality control, and industrial production practices.

- Developed practical understanding of the relationship between **design, manufacturing, inspection, and assembly**.

CERTIFICATIONS

Product Design and Manufacturing – NPTEL	2026
SolidWorks – GUVI	2026
ANSYS Fluent (CFD Simulation) – GUVI	2026
Advanced Robotics – NPTEL	2025
ROS 2 Humble Workshop – MIT	2024

ACHIEVEMENTS

Top 15 Finalist – Autodesk India Design Week (Selected among 200+ teams for the national 24-hour design hackathon)	
2026	
Third Prize – CAD Design Competition	2025

ACTIVITIES LEADERSHIP

Vice Captain – SAE DDC Fixed-Wing UAV Team
SAE India Member
NCC Naval Wing Cadet
Participated in national-level product design, CAD, drone, and engineering competitions.